

Project Proposal (Group 5): EV Charging Infrastructure & Adoption Analytics

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Abstract

As electric vehicle adoption accelerates across the United States, a critical question emerges: is the charging infrastructure keeping pace? This project builds an end-to-end data analytics pipeline to analyze the volume and geographic density of EV charging stations across US cities and states, and cross-references that data against state-level EV registration counts to surface infrastructure gaps. We combine three data sources: the **NREL Alternative Fuels Station API (real-time, pulled daily)** for current station inventory, **the DOE/AFDC historical EV registration dataset (archive, 2016-present)** for state-level adoption trends, and the **US Census Bureau ACS5 API** for state population figures used to normalize metrics on a per-capita basis. Data ingestion will be orchestrated using **Apache Airflow** and loaded into **Snowflake (RAW schema)**, cleaned and standardized via **dbt (CURATED schema)**, and aggregated into **analytical models (ANALYTICS schema)** that power a **Preset.io dashboard**. Key metrics would include:

- EV Station Density Map
- Station Count by State and city
- Top 20 Cities
- EV Adoption vs Infrastructure
- Infrastructure Gap Ranking
- Stations per 100,000 residents

In addition to descriptive analytics, the project will implement a time-series forecasting model to predict EV adoption growth and charging infrastructure expansion over the next 12 months, enabling proactive identification of high-risk infrastructure gap states

Possible Stakeholders

Dashboard is designed for:

- State transportation agencies
 - EV charging network operators
 - Energy planners seeking to prioritize infrastructure investments and forecast charging demand growth.
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Dataset Links and Descriptions

Dataset 1: NREL Alternative Fuels Station API (Real-time)

- **Link:** <https://developer.nrel.gov/docs/transportation/alt-fuel-stations-v1/all/>
- **Registration:** Free API key at <https://developer.nrel.gov/signup/>
- **Description:** Maintained by the US Department of Energy's National Renewable Energy Laboratory, this API provides a live, regularly updated inventory of all alternative fuel stations in the US. Filtered to fuel_type=ELEC, it returns ~65,000+ EV charging locations with fields including station name, city, state, ZIP, coordinates, connector types, Level 1/Level 2/DC Fast Charger port counts, operational status, and open date. The underlying dataset is refreshed daily by NREL networked stations (ChargePoint, EVgo, Blink, etc.) are auto-imported from charging network partners, while non-networked stations are maintained manually. A dedicated last-updated endpoint (/v1/last-updated.json) allows the pipeline to check whether data has actually changed before triggering a full pull. This serves as the **real-time data source**, pulled daily via Airflow.

Dataset 2: DOE/AFDC State EV Registration Data (Historical Archive)

- **Preferred option** - Link: <https://afdc.energy.gov/vehicle-registration>
- **Backup option:** <https://www.atlasevhub.com/materials/state-ev-registration-data/>
- **Description:** Published by the US Department of Energy's Alternative Fuels Data Center, this dataset contains annual EV registration counts broken down by state and vehicle type (BEV and PHEV) from 2016 to present. It is available as downloadable CSVs and provides the historical longitudinal view of EV adoption across all 50 states. This serves as the **historical archive source**, loaded once and refreshed annually.

Dataset 3: US Census Bureau ACS5 API (Population Reference)

- **Link:** https://api.census.gov/data/2024}/acs/acs5?get=NAME,B01003_001E&for=state:*
 - **Registration:** No API key required for basic access
 - **Description:** The American Community Survey 5-Year Estimates from the US Census Bureau provides state-level population figures. This dataset is used exclusively as a denominator to compute per-capita density metrics (e.g., stations per 100k residents, EVs per 100k residents), enabling fair comparisons across states of different sizes.
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Reference Links

- NREL AFDC Developer Docs: <https://developer.nrel.gov/docs/transportation/alt-fuel-stations-v1/>
- DOE EV Registrations by State: <https://afdc.energy.gov/vehicle-registration>
- US Census Bureau API Guide: <https://www.census.gov/data/developers/guidance/api-user-guide.html>